UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2024

CSAT : PAPER-II

(Analysis with Explanation)

Time : 2 hours **(Exam date : 23.06.2024)** **Maximum Number : 200**

Directions for the following 4 (four) items:
Read the following two passages and anwer the items that follow the passage. Your answer to these items should be based on the passages only.

Passage–1

''The history of science is the real history of mankind.'' In this striking epigram, a nineteenth-century writer links sciene with its background. Like most epigrams, its power lies in emphasizing by contrast an aspect of truth which may be easily overlooked. In this case, it is easy to overlook the relations between science and mankind, and to treat the former as some abstract third party, which can sometimes be praised for its beneficial influences, but frequently and coneveniently blamed for the horrors of war. Science and mankind cannot be divorced from time to time at men's convenience. Yet we have seen that, in spite of countless opportunities of improvement, the opening years of the present period of civilization have been dominated by international conflict. Is this the inevitable result of the progress of science or does the fault lie elsewhere?

1. Which of the following is/are emphatically conveyed by the author of the passage?
1. Without science, mankind could not have continued to exist till today.
2. It is the science that will ultimately determine the destiny of mankind.
Select the correct answer using the code given below.
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Options 1 is not correct, because the statement "without science, mankind could not have convinced to exist till today" is impractical and inappropriate according to the passage. The passage does not stated the existence of mankind is only because of science. Science can make our life easy and comfortable.
• **Options 2 is also not correct,** because the passage does not claims science will determine mankind's destiny. It clarify the role of science on social aspects. The relationship between science and mankind's destiny is comply and possible influenced by other factors. So, the option 2 is also not appropriate according to the passage.

2. Based on the above passage, the following assumptions have been made:
1. The horrors of modern life are the inevitable result of the progress of science.
2. The aspect of truth likely to be overlooked is that science is what man has made it.
Which of the assumptions given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b)
• **Assumption 1 is not valid,** because the statement "the horrors of modern life are the inevitable result of the progress of science" is not appropriate according to the passage. It contradicts the assertion that the horrors of modern life are the inevitable result of the progress of science, suggesting that the real cause lies elsewhere.
• **Assumption 2 is valid,** because the passage discusses aspect of truth likely to be overlooked is that science is what man has made. It means according to need people changes the difinition of science and uses it in various aspects. This passage focusses well with the inter connectedness of science and human agency.

Passage–2

Only with long experience and opening of his wares on many a beach where his language is not spoken, will the merchant come to know the worth of what he carries, and what is parochial and what is universal in his choice. Such delicate goods as justice, love and honour, courtesy, and indeed all the things we care for, are valid everywhere but they are variously moulded and often differently handled, and sometimes nearly unrecognizable if you meet then in a foreign land, and the art of learning fundamental common values is perhaps the greatest gain of travel to those who wish to live at ease among their fellows.

3. When we meet other people while we travel, we learn to diferentiable between:
(a) Imagination and understanding
(b) Communities and nationalities
(c) Local values and universal values
(d) Friends and foes

Ans. (c) : Options (a) is not correct, because the passage does not differentiate between imagination and understanding. The passage discusses the universal values and local understanding. It focuses, how to understand better through travel and interaction. So, this statement imagination and understanding' does not appropriate according to the passage.
• **Option 'b' is not correct,** the passage discusses about various culture and language. It does not differentiate between communities and nationalities. It is about to understand the essence of common values, irrespective of geographical and & cultural boundaries.
• **Options 'c' is correct,** "The statement "Local values and universal values" is appropriate according to the passage. Justice, love and honor, courtesy and indeed all the thing we care for all are called universal values.
• When someone meet the people in a foreign land, he learn different common values at local level.
• **Option 'd' is not correct,** because the passage does not focuses about differentiating between friends and foes during travel. This option is not under the scope of the passage.

4. With reference to the above passage, the following assumptions have been made:
1. Travel leads to an understanding of humans.
2. Travel helps those who wish to learn fundamental common values.
3. A person with long experience in travel can resolve differences amongst people.

Which of the assumptions given above are valid?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (a) : • Assumption 1 is valid, because the statement "Travel leads to and understanding of human" is perfectly correct according to passage. The passage defines that a person during travel learns worth of what he carries, including values like justice, love and honour. So in this way a person learn different human values and behaviors across different cultures.
• **Assumptions 2 is also valid,** because during travel a person learn fundamental common values, that is great gain of travel. So, this assumption directly supports the passage.
• **Assumptions 3 is not valid,** The passage discusses that a person becomes knowledgeable and accustomed about different values. He learn everything around the world but it is not guaranteed that he can mediate or resolve conflicts.

5. Let X be a two-digit number and Y be another two-digital numbers formed by interchanging the digits of X. If (X + Y) is the greatest two-digit number, then what is the number of possible values of X?
(a) 2 (b) 4 (c) 6 (d) 8

Ans. (d) : According to given conditions of in the question– x + y = 99
(10T + U) + (10U + T) = 99
11T + 11U = 99
T + U = 9
Possible values for T + U = 9, are (1, 8) (8, 1) (2, 7), (7, 2), (3, 6), (6, 3) (4, 5) and (5, 4)
Hence for x there are eight values which will be possible values.

6. Consider the following:
Weight of 6 boys = Weight of 7 girls = Weight of 3 men = Weight of 4 women
If the average weight of the women is 63 kg, then what is the average weight of the boys?
(a) 40 kg (b) 42 kg
(c) 45 kg (d) 63 kg

Ans. (b) : Average weight of women is = 63 kg.
we can write, weight of 6 boys = weight of 4 women.
= 4 × 63 kg.
6 boys = 4 × 63 kg.
1 boys, weight = $\frac{4\times63}{6} = 42\text{kg}$
This will be average weight of boys.

7. How many times the hour hand and the minute hand coincide in a clock between 10: 00 a.m. and 2:00 p.m. (same day)?
(a) 3 times (b) 4 times
(c) 5 times (d) 6 times

Ans. (a) : Hour hand and minute hand coincide in 1 hr.
10AM – 11AM ⇒ 1 time coincide
11AM – 12 PM ⇒ 1 time
12 PM – 1 PM ⇒ 0 {in this time period no coincidence occours}
1PM – 2PM ⇒ 1 time
There are total 3 times coincidence will occurs between 10 : 00 am to 2 : 00 pm.

8. The calendar for the year 2025 is same for:
(a) 2029 (b) 2030
(c) 2031 (d) 2033

Ans. (c) : Ticks:

Years		Repetition after years.
Leap year	:	28
Leap year + 1	:	6
Leap year + 2	:	11
Leap year + 3	:	11

Years 2025 can be written as leap year 2024 + 1 year.
So, some calendar will be in (2025 + 6 = 2031).

9. Let p, q, r and s be distinct positive integers.
Let p, q be odd and r, s be even.
Consider the following statements.
1. $(p - r)^2$ (qs) is even.
2. (q – s) q^2s is even.
3. $(q + r)^2$ (p + s) is odd.
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : As we know,
→ odd number × even number = even number
→ odd number + even number = odd number
There is given that–
p, q are odd numbers – and r, s are even numbers.
• $(p - r)^2$ (q, s) is even, it is correct because (s) is even number.
• (q – s) q^2s is even, it is also correct because s is even number .
• $(q + r)^2$ (p + s) is odd, if is also correct, because P is odd no. + S is even no ⇒ odd number.
Some of odd & even number will be odd number.
Hence all statements are correct.

10. What is the angle between the minute hand and hour hand when the clock shows 4:25 hours?
(a) 12.5° (b) 15°
(c) 17.5° (d) 20°

Ans. (c) : Angle between the minute hand and hour hand
⇒ $Q = \frac{11}{2}M - 30H$
Where Q = angle between minute & hour hand.
M = minutes
H = hours
So, $Q = \frac{11}{2}\times25 - 30\times4$
$\boxed{Q = 17.5^\circ}$

Directions for the following 3 (three) items:
Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage–1

By the time children reach class 8, the bulk of them tend to be in the age range of 13 years to 15 years. But in our country, about a quarter of all children in class 8 struggle with reading simple texts and more than half are still unable to do basic arithmetic operations like division. Every year about 25 million young boys and girls form elementary school move into the life that lies for them beyond compulsory schooling. They cannot enter the workforce at least in the organized sector till they are 18. For many families, these children are the first from their families ever to get this far in school. Parents and children expect that such 'graduates' from school will go on to high school and college. Hardly anyone wants to go back to agriculture. On the other hand, abilities in terms of academic competencies are far lower than they should be even based on curriculum expectations of class 8.